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Project Title	FYP25CM009: Simulation of a track car		

1. Project Objectives

The program will be designed to determine an ideal racing line around a circuit by applying the fundamental laws of motion and vehicle dynamics. The development process will be structured in progressive stages, beginning with a simple circular track to verify core functionality and advancing toward more complex track geometries. For each circuit configuration, the system will generate an overlay of the computed optimal line, estimate the maximum achievable velocity at each point around the track, and calculate the predicted lap time based on vehicle parameters, including tyre grip, acceleration capability, and braking performance.

2. Description of work completed

2.1. Create a track using Splines from Waypoints

The primary objective of the project is to utilise splines to create a track using waypoints defined within predefined parameters. The spline maps a path between the points, completing three separate loops, with the middle loop (the second) being the most defined. The loop will have N amounts of points with equal distance between them. From this, I can then synchronise the length of the digital track to the final track length. This can be seen in Appendix A1

2.2. Work out max point speed for N points.

The max speeds around the track, which will be the cornerstone of any lap times, are determined by the Coefficient of friction μ , which gives the max speed of the corner when put into the formula $v = \mu \cdot g \cdot r$, where v is the maximum velocity, g is the force of gravity, and r is the radius of the corner. This can be seen in Appendix A.2

$$v = \sqrt{\mu g r} \quad (\text{F.1})$$

2.3. Using max speed points, work out the lap time.

Once the max speeds are worked out, we can get a very basic lap time by accelerating the car from the start line until reaching a point N. Upon reaching N, if the car's velocity is greater than the max speed, set the speed equal to the max speed and continue accelerating. These speeds can be computed using the three primary equations of motion. This gives a basic lap time that will have higher accuracy on tracks with a low number of braking events. The traces given by the program can be seen in A.3 and A.4

$$v = u + a * t \quad (F.1)$$

$$s = u * t + \frac{1}{2} * a * t^2 \quad (F.2)$$

$$v^2 = u^2 + 2 * a * s \quad (F.3)$$

3. Evidence of work completed

So far, up to and including the work described in (2.3), has been completed. The singular line of the track has been computed using a spline of n points. Additionally, the max speed points have been calculated and recorded using the coefficient of grip formula (F.1). This data was then used to create a basic lap time using the motion formulae (F.2, F.3, F.4) and instantaneous braking initially.

4. Literature review

4.1. Book: How to Win a Grand Prix by Bernie Collins (Collins, 2025)

In Collins' book, she describes the software used within strategy teams in F1, which seems to be based on raw data from the track rather than physics-based simulations. Collins also referenced how they can run these simulations hundreds of times, showing how the car may change lines based on tyres and conditions to gather useful information for upcoming races.

4.2. Book: Real World Physics by Dan O'Regan (O'Regan, 2000)

Real World Physics by O'Regan contains all the physics information that is needed for this project to supplement the information given by my mentor, Charles Markham. It goes into detail on the laws of motion that will be key to the programming of my project. It is a sufficient and valuable resource for the project.

4.3. Web Page: Processing Reference Page (*Processing Reference Page*, 2025)

The Processing reference page has been the main resource for creating graphics elements for the project, such as the spline, speeds, and timing

5. Use of GenAI and tools

5.1. Grammarly

Grammarly is an assistant for report writing. When struggling with grammar and spelling, Grammarly lets writers focus on conveying ideas and arguments without worrying about their shortcomings in writing. It also allows for the tone of the document to be analysed; this allows those who may struggle with academic writing to be notified if they are being too informal.

5.2. ChatGPT

ChatGPT (a large language model) can be used to help reformat code or to give advice when a program is not giving an expected output. ChatGPT was used in this project to create a method that would output the data to an Excel sheet, allowing the visualisation of data in a usable manner. When dealing with unfamiliar contexts, it can be a very useful companion.

6. Future Work

6.1. Integrate a smarter braking solution.

The braking should be based on the motion formulae and not instantaneous slowing alone. The project should aim to integrate this into future updates to get a more realistic and accurate lap time.

6.2. Racing Line around track.

Once this is completed, the track should be expanded to have three separate lines. This will allow for a machine learning element, allowing the car to choose which line to take to get around the track as fast as possible, based on grip and the v-max at each point.

6.3. Future Expansion.

Looking to the future, the project can be expanded to have multiple lanes with many options for a machine learning element to learn from. Upon this being completed, the racing lines that will be chosen will be more intelligent and realistic.

Also, the integration of a dynamic element to the car would allow for tyre degradation; the inclusion of different car acceleration, braking abilities, fuel burn and mass changes. There is already a tyre model built into the program, so setting up the program to run a complete race of 'n'3 laps and 'm' pitstops would be possible.

References

Collins, B. (2025) *How to Win a Grand Prix: From Pit Lane to Podium the Inside Track on F1*. illustrated edn. Hachette Australia, 2024.

O'Regan, D. (2000) *Real World Physics*. Folens Publishers. Available at:
<https://studyclix.blob.core.windows.net/static/content/file/uploads/d/d4a3e6b7-80ec-41e1-b128-bd6d38f76257.pdf>.

Processing Reference Page (2025). Available at: <https://processing.org/reference>.

Appendices

A.1 Editing Waypoints of Brands Hatch Indy

0:37.71

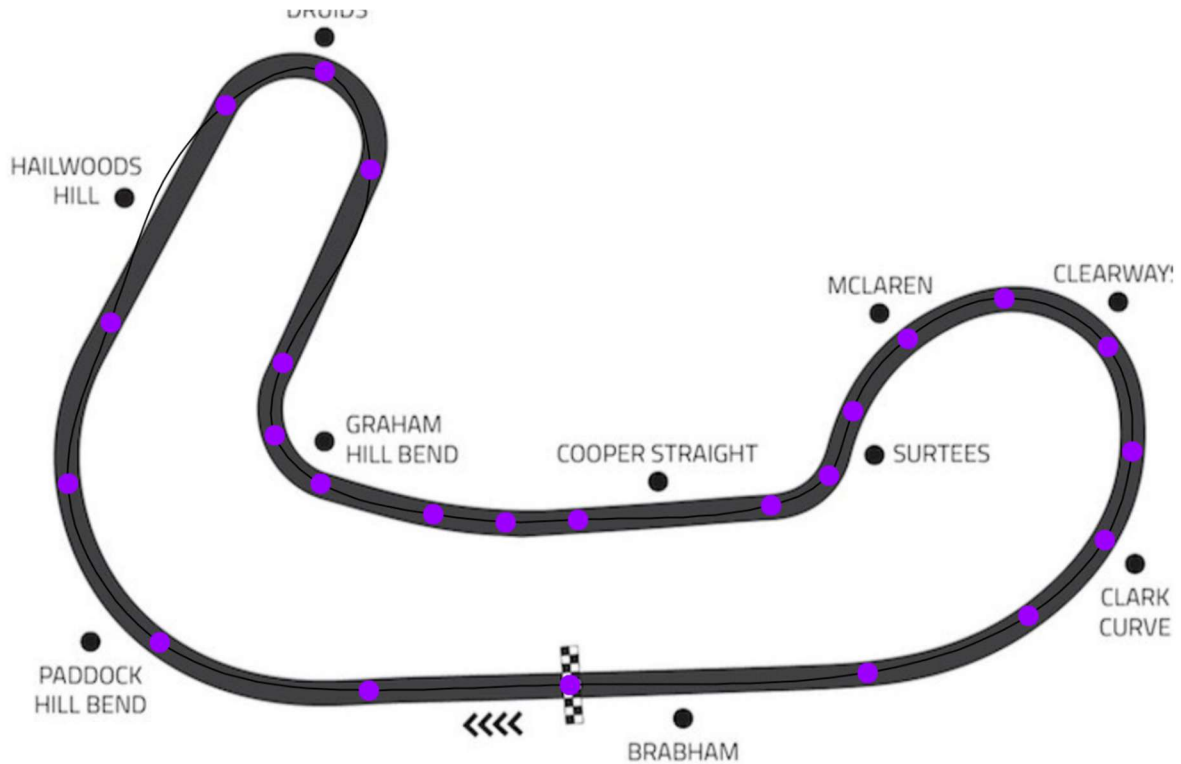


Figure A.2 Speed trace available for Brands Hatch Indy in Excel

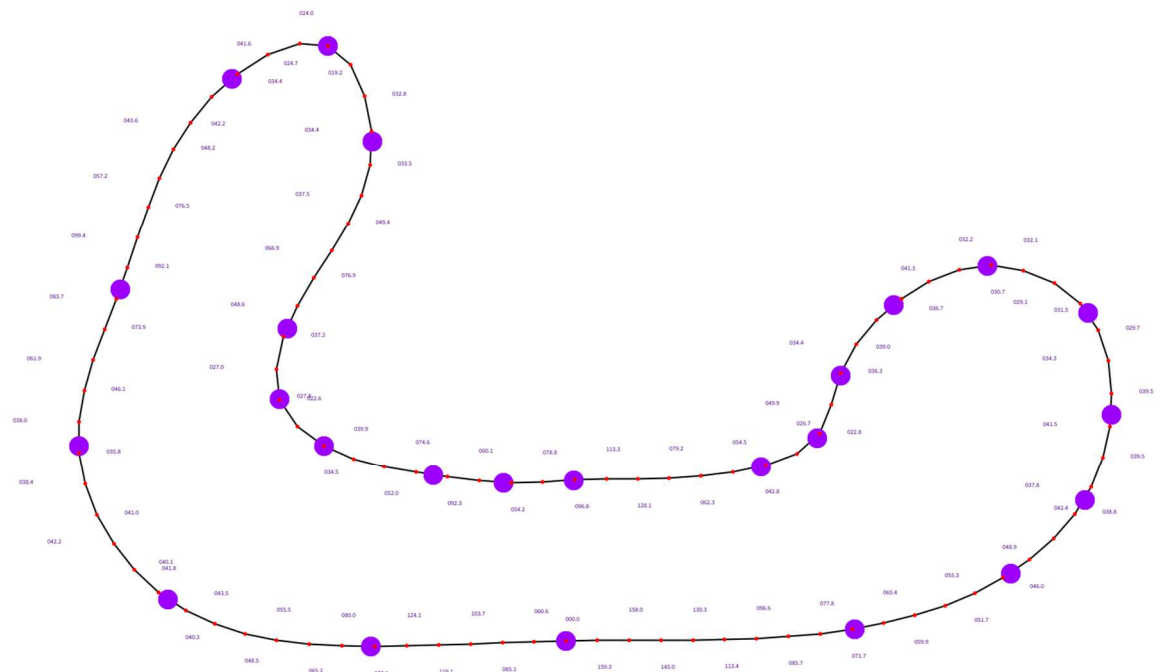


Figure A.3 Speed trace available for Brands Hatch Indy in Excel

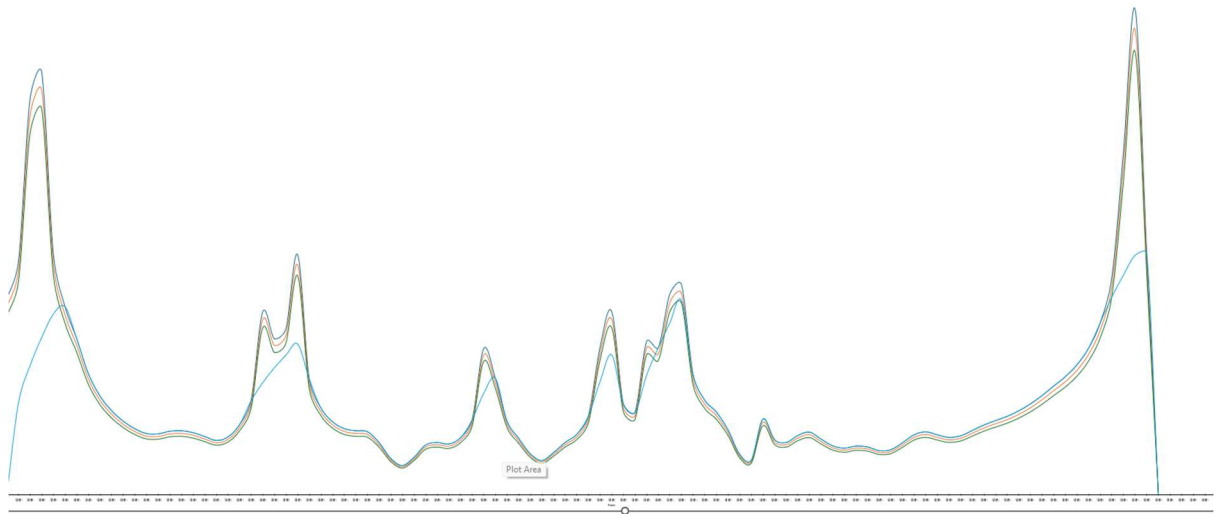


Figure A.4 Speed trace available for Brands Hatch Indy in Excel (zoomed)

